Client: ShopSight
Background Research

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Team Statement:
By submitting this background document, the members of our team affirm that we all participated equally in the research for and the writing of this document.

Statement of the Design Challenge

Shopsight faces the strategic challenge of transitioning from a web-based platform to a mobile first application. They have successfully established a co-creation platform where customers can participate in design challenges and trend prediction that generates millions of product images daily. They wish to prioritize features from the existing web application that would belong in the first version of the app like Design Challenges, AI Agents, Trend shopping, Future pulse reports and Rewards. Simultaneously, also exploring new mobile specific features such as Personalized AI shoppers, sustainability dashboards, collaboration boards, social shopping and eco-badges which would make the most impact. Success for ShopSight means building a mobile-first platform that becomes a daily destination for Gen Z creativity, delivers real-time consumer insights that help brand partners reduce product waste, and demonstrates measurable sustainability impact that strengthens the company's appeal to investors and stakeholders.

Problem Context

Traditional brand research that relies on focus groups, lengthy surveys, customer segmentation and user personas has been a hit and miss for companies in the past. Brands feel that they connect with the customers and their needs but in reality, the retail industry has struggled with massive waste generation with stacks and stacks of unused and unsold inventory along with failed product launches costing them billions of dollars. Research from Future Earth (2024) reveals that retailers globally still grapple with an excess inventory rate exceeding 30%, with unsold merchandise accounting for approximately 17-20% of total inventory.

The evolution to consumer co-creation platforms reflects a massive shift from passive market research to consumers being actively involved in product development, research, and feedback. Nguyen et al. (2024) reveals that value co creation significantly influences customer engagement and loyalty, with customers' participation in design processes amplifying satisfaction rates by 38%. This transition has been accelerated by technological advancements that enable real time feedback collection with platforms achieving 22 second average response times and 85-90% prediction accuracy as documented in Shopsight's operational data.

Mobile app feature prioritization is complex process weighing user engagement against genuine sustainability communication. While 95% of consumer goods companies make innovation a core company strategy, apps have to choose between "must-have" and "nice-to-have" features based on models such as RICE scoring and MoSCoW matrices. Sustainability-oriented platforms have the added complexity of measuring environmental impact in conjunction with user engagement metrics.

The sustainability communication gap presents immediate challenges to mobile platforms. With more and more people anticipating increased sustainability communication, over half of them feel that corporate pledges are impossible since there is much greenwashing (Sajid et al., 2024; McKinsey, 2023). This skepticism particularly challenges mobile apps in gamifying environmental impact, as people demand real, measurable benefits and not superficial eco-messaging.

Gen Z sustainability consumption habits reveal contrasting tendencies making platform design decisions challenging. Recent surveys reveal sustainability somewhat low on the agenda for Gen Z, yet environmental serious games are successful in facilitating sustainable action fulfillment when addressed with genuine measurement of impact (Opeepl Research, 2024; Chen et al., 2025). The attitude-behavior gap persists, with the majority willing to pay for green products but few actively taking the effort to search for them upon purchase.

This context makes ShopSight's feature prioritization challenge particularly relevant, considering the website has to juggle conflicting agendas of engaging mobile experiences and genuine sustainability focus while providing value for consumers and brand partners within an increasingly cynical market environment.

Stakeholders

The transition to a mobile-first platform for ShopSight will affect a wide range of stakeholders. At the center are Gen Z users who are deeply engaged in fashion culture and digital trend participation. They experiment with style, follow emerging aesthetics, and use mobile platforms for creative expression. These users contribute both content and validation by generating ideas, sharing and evaluating contributions, and influencing what becomes a marketable trend. Micro-influencers form a particularly valuable segment within this group. On TikTok and Instagram they already drive viral fashion challenges, thrift flips, and outfit inspiration that spread quickly through their loyal audiences, making them critical to ShopSight's adoption. Their expectations for seamless mobile functionality, visually engaging design, and impact-driven incentives will shape which features sustain long-term use. Key functionality such as gamified rewards, AI-based personal shopping agents, and collaboration boards must align with these expectations to ensure the platform delivers clear value.

Brand partners represent another vital stakeholder group. These include both emerging sustainable fashion labels and established retailers seeking to remain relevant with Gen Z audiences. Their interest in ShopSight lies in accessing trend data early, validating product ideas through consumer co-creation, and aligning assortments with fast-shifting preferences. To meet these needs, the platform must deliver insights that are both timely and actionable, while clearly demonstrating that user engagement translates into market relevance. Internal stakeholders are equally significant, especially ShopSight's product, engineering, and design teams. They hold the responsibility for transforming the mobile-first strategy into a scalable, intuitive, and visually compelling application. Their decisions about which features to prioritize, how to schedule development, and what trade-offs to make in user experience will directly determine feasibility, quality, and long-term impact.

To deepen our understanding of these stakeholder groups, we will conduct both interviews and direct observation. Our interview process will include one-on-one conversations with Gen Z consumers across different user personas, such as trendsetters, casual co-creators, and sustainability-conscious shoppers. We will also interview fashion brand marketers, product designers, and retail strategy professionals to understand what types of insights, data visualizations, or content integrations would be most valuable in a B2B context. On the observational side, we will analyze user behavior on web based co creation platforms such as Threadless, Redbubble, and DesignCrowd, focusing on how users submit designs, vote on content, and interact with community features. Additionally, we will observe mobile shopping behaviors on Depop, Instagram Shop, and TikTok Shop, noting friction points, patterns of discovery, and moments of user delight. This combination of qualitative interviews and ethnographic observation will enable us to map stakeholder journeys, identify unmet needs, and validate which mobile features will provide value across the ShopSight ecosystem.

Competition

The fashion industry moves at a relentless pace, with trends sparked by digital culture and fading almost as quickly. To anticipate demand, companies lean on forecasting agencies, market researchers, and social media analytics. These approaches provide structure, but they are expensive, time-consuming, and

fundamentally reactive. As Koren and Shnaiderman (2023) explain in their study of forecasting and supply-chain costs, volatility and short product lifecycles make it almost impossible for brands to perfectly align production with consumer demand. They show that statistical forecasting models, such as ARIMA-based time series methods, can reduce costs by minimizing inventory errors. However, these models are rooted in historical sales data, which cannot keep pace with rapidly shifting consumer tastes. This research highlights a key limitation, that even the most mathematically sophisticated models' risk being outdated by the time products reach the market. The challenge does not stop at finances but more importantly environmental as well, as overproduction leads directly to unsold inventory and waste (Koren & Shnaiderman, 2023). What is missing is a way to harness consumer creativity in real time and transform it into actionable insight before products are made, a gap that ShopSight is positioned to fill.

Over the past decade, different types of platforms have tried to fill the insight gap, each approaching the problem from its own angle. Forecasting firms continue to rely on experts who interpret cultural signals into style predictions meant to guide upcoming collections. Artificial intelligence has been introduced to make forecasting faster, with startups scanning social media and online images to detect emerging patterns. Shi et al. (2021) explore these methods in detail, showing how AI models such as Faster R-CNN and CNN classifiers can extract clothing attributes, analyze runway collections, and summarize fashion trends at scale. Their study demonstrates the efficiency and objectivity of AI-driven forecasting, which reduces human bias and speeds up data processing. Yet the authors also stress that AI remains reactive and can only identify signals already present in visual datasets, whether on social media or e-commerce platforms. This means that AI-enabled competitors are still describing the present and recent past rather than predicting the future (Shi et al., 2021). On the retail side, analytics platforms focus on monitoring competitors and adjusting pricing or assortments as markets shift. A smaller number of communities have experimented with crowdsourcing, inviting consumers to submit and vote on designs, while sustainability organizations work to simplify complex supply-chain data into accessible ratings for shoppers (Wallpaper*, 2022). These approaches highlight the ingenuity of the industry, but because they operate in isolation, they leave brands piecing together partial answers. What is still missing is a single solution that can capture real-time consumer creativity, validate it through predictive models, and embed sustainability into decision-making from the outset; the exact integration that ShopSight proposes to deliver.

Five competitors in particular highlight the diversity of current solutions- *WGSN, Heuritech, EDITED, Threadless,* and *Good On You.* Each offers value to brands, shoppers, and investors, yet each also reveals blind spots that prevent companies from uniting forecasting, consumer engagement, and sustainability in a coherent way.

WGSN is the most established name in trend forecasting, translating cultural movements into detailed design directions that influence choices on colors, fabrics, and styles. Teams across fashion, retail, and interior design use WGSN's reports as a shared reference, giving them a structured way to reduce failed launches and align strategy. Its strength lies in authority and consistency, with a track record that makes it highly respected by both brands and investors (WGSN, 2024). Yet as Koren and Shnaiderman (2023) argue, even the most advanced forecasting models tend to focus on minimizing supply chain costs rather than enabling new forms of consumer-driven creativity. This exposes a structural weakness in WGSN's expert-driven model; while it provides confidence and stability, it is not designed for agility. In an environment where micro-trends emerge from TikTok or Instagram and disappear in weeks, WGSN's long-horizon approach often leaves brands struggling to react in time (Vogue Business, 2023).

Heuritech uses artificial intelligence to address the speed gap by analyzing millions of images shared daily on social media. Its algorithms detect the rise of specific silhouettes, patterns, or accessories and project how these might spread across markets and demographics (Heuritech, 2024a). Brands value the precision of these insights for refining assortments and managing inventory, and case studies show companies like *Mango* applying its forecasts to reduce overproduction (Heuritech, 2024b). Shi et al. (2021) provide strong evidence for this approach, demonstrating that AI-powered image recognition can process data far faster than human forecasters and reduce subjectivity in trend analysis. However, they also emphasize that these methods rely on datasets drawn from what consumers are already wearing or what designers have already produced. In other words, Heuritech and similar platforms excel at mapping current adoption but cannot generate forward-looking creativity. This leaves them as sophisticated but reactive tools, useful for optimization but limited in shaping the future.

EDITED focuses on operational intelligence rather than cultural forecasting. Its platform aggregates competitor catalogs, pricing updates, and sell-through data, giving retailers a live dashboard of market movements (EDITED, 2024). For buyers and merchandisers, this information is critical for adjusting price points, spotting gaps in assortments, or planning promotions. The platform is especially respected for its depth of retail data and speed of updates (Business of Fashion, 2021). Yet EDITED's strength in optimization is also its weakness. It enables retailers to react quickly within the existing system but does not offer new insight into what consumers might want in the future or how to align assortments with sustainability goals.

Threadless represents a participatory alternative by actively involving consumers in the design process. Community members submit T-shirt graphics, vote on favorites, and see selected concepts produced and sold. This model builds strong loyalty among engaged users and reduces guesswork for the company by validating designs before production (Threadless, 2023). It proves that consumers are willing to co-create rather than only consume. However, Threadless operates at a limited scale and scope, focused on graphic apparel rather than the full spectrum of fashion categories. Its insights remain community-specific and do not provide predictive analytics or industry-wide application, leaving it more of a niche experiment than a comprehensive forecasting solution.

Good On You addresses fashion's sustainability problem by rating brands on their treatment of workers, use of resources, and environmental impact. Consumers use the ratings to quickly assess whether a label aligns with their values, and retailers often highlight thdeese scores in "conscious edit" collections. For investors, the ratings provide an independent benchmark of ESG practices (Good On You, 2024). The strength of Good On You lies in its independence and transparency, which helps cut through the greenwashing common in fashion marketing (Vogue Business, 2023). Its limitation is that the ratings are retrospective. They assess current practices but do not shape what kinds of products brands will design or how sustainability can be built into new launches.

Taken together, these five competitors show how scattered the industry's solutions remain. WGSN delivers structure, Heuritech offers speed, EDITED ensures efficiency, Threadless experiments with participation, and Good On You enforces accountability. Yet the detailed findings of Koren and Shnaiderman (2023) and Shi et al. (2021) underline why these competitors ultimately fall short. Forecasting models reduce errors but are tied to past data. AI accelerates recognition but only after trends are already visible. None connect creativity, prediction, and sustainability into one seamless process. This is precisely where ShopSight stands

apart. By turning consumer ideas directly into predictive insights and embedding sustainability storytelling at the heart of the process, ShopSight can close the loop that competitors leave open and give brands a timely, accurate, and responsible way to innovate.

Market Trends and Industry Insights

Recent studies on consumer behavior, sustainability, and mobile commerce provide valuable insights into the key market trends shaping the retail industry and highlight important ShopSight's mobile app strategy. According to Strategic Direction (2022), it is evident that retailers often segment customers based on demographic factors like gender. However, both genders seem to be more concerned about their overall purchasing experience and satisfaction with an online environment than trusting a website enough to make an online purchase.

Similarly, Castro-López, Iglesias, & Puente (2021) confirm that there is a change in consumer consumption habits. The fashion industry is considered one of the most polluting industries in the world, after the oil industry, being responsible for 10% of the production of global CO2 emissions and 20% of global water waste. As per Breznikar report, In the U.S., the average person throws away about 36-37 kg of used clothing per year. Many garments (like cotton shirts) need thousands of liters of water to produce. These habits contribute substantially to water usage, CO_2 emissions, and landfill waste. There is a high increase in Consumers who are bending towards sustainable fashion as a tool to preserve the environment and the health & well-being of society and are also willing to pay more for a slow fashion product. An increasing number of consumers stick to this new trend that focuses on slowing down the production and consumption processes and promoting sustainable values among all actors involved in the fashion system as a path to improve the environment and the health of society.

In addition, Andronie Mihai and others (2021) identifies several consumer motivational factors influence the use of mobile shopping applications:

1. Online buyers are more likely to prefer purchasing products through retailer desktop websites instead of mobile websites on smartphones.

2. The spread of digital technologies, combined with customer's intentions to adopt smart retail innovations is rapidly expanding in physical stores due to cutting-edge developments.

3. The growing popularity of online social networks affects the shopping decisions of mobile app users on social commerce platforms in a significant percentage.

4. For users with lower levels of compatibility belief, mobile shopping loyalty is mostly driven by their perception of vendor trust.

Putting all this together, these trends have both near-term and long-term impacts. In the near term, adoption of a new shopping app may be gradual as many consumers still prefer desktop but Gen Z's comfort with mobile and social shopping makes them a key early audience. Sustainability is already acting as a differentiator today with many consumers expressing willingness to pay more for eco-friendly products. However, in the long term, sustainable consumption is more likely to evolve from a consumer preference to a baseline expectation particularly among younger generation. Similarly, AI-driven personalization and predictive insights will likely become standard components of mobile retail, meaning features such as AI shoppers and sustainability dashboards will be essential for ShopSight's positioning in the market.

These industry shifts also create distinct opportunities and challenges for ShopSight. Opportunities are positioning sustainability as a competitive advantage, using social shopping and collaboration features to

build community engagement and using AI-driven insights to help brands reduce waste and improve decision making. ShopSight can build brand and win consumer's trust by integrating transparent sustainability features such as eco badges, impact dashboards and eco-scores *(Anja Weber, 2021)* that rate products on their environmental and social footprint. Eco-scores are becoming more widely recognized in many countries and following them would allow consumers to compare products at a glance, reducing confusion and addressing concerns about greenwashing. By making the use of collaboration boards and social shopping features it can capitalize on the growing influence of social networks on purchase decisions. Highlighting AI as a driver of both personalization and waste reduction positions ShopSight as an innovation leader in sustainable retail. However, challenges remain are adoption friction may slow early growth if consumers hesitate to move away from desktop websites, greenwashing concerns make transparency and credible reporting essential and prioritizing the right mix of features will be critical to avoid overwhelming new users.

By recognizing these trends and its impacts, ShopSight can strategically prioritize features that deliver immediate value such as user-friendly design, transparent sustainability features on their mobile app, and early social shopping tools to its customers while simultaneously building toward long-term differentiation with advanced AI personalization and predictive insights.

Key Insights

Our research shows that the fashion insight space is still very split. Some platforms only look at forecasting, others only track retail data, and a few focus mainly on sustainability. What is missing is a place that brings these pieces together in one mobile experience. Companies like WGSN, Heuritech, and EDITED are strong in their areas, but none of them take consumer creativity in real time and turn it into useful insights that also push sustainability forward. That gap gives ShopSight the chance to stand out by letting Gen Z not just follow trends but actually shape them.

When it comes to Gen Z, the story is complicated. They often talk about caring for the planet, but in practice things like convenience, style, and personal fit tend to win. This means sustainability cannot sit on the side. It needs to be blended into the app in ways that feel natural. Gamified rewards, community challenges, and clear signals of impact are ways to do that. These users are not passive either. They like to create, criticize, and be heard. ShopSight will need to give them real ways to contribute and see their ideas matter.

Current market shifts make this even clearer. Social commerce is changing how people shop. Platforms such as TikTok Shop and Depop show that people buy into stories and community more than a standard shopping funnel. For ShopSight, features like collaboration boards, AI shoppers, and trend challenges should be designed as creative tools, not just utilities. Brands are also looking for something more than engagement numbers. They want insights that are fast, trustworthy, and connected to sustainability goals.

All of this points to the same direction. ShopSight should build mobile features that feel quick, expressive, and social for users, while giving brands reliable data they can act on. These insights will guide which features we put forward, how we design prototypes, and what we test with users. The aim is not only to be innovative but also to respond to what the market and its stakeholders really need.

Appendix A

1. “How does Heuritech use artificial intelligence to predict fashion trends?”
2. “What are EDITED’s main features as a retail analytics platform?”
3. “What is Good On You, and how does it evaluate brands for sustainability?”
4. “Retrieve global retail waste and excess inventory statistics for 2024–2025, including percentages and cost impacts”.
5. “Identify peer-reviewed studies on consumer co-creation in retail and best practices for mobile app feature design”.
6. “Summarize research on Gen Z attitudes and behaviors toward sustainable consumption, citing sample sizes and key findings”.
7. “Find experimental studies on gamification for environmental behavior change, including effect sizes and engagement metrics”.
8. “Investigate scholarly analyses of sustainability communication challenges, focusing on consumer skepticism and greenwashing impacts”.
9. “List and compare product feature prioritization frameworks (e.g., RICE, MoSCoW), including criteria, scoring methods, and case study examples”.
10. “Where does fashion industry stand in global CO_2 emissions and water waste”?
11. “How do social networks influence purchase decisions in mobile shopping”?
12. “What are eco-scores and how do they influence consumer decision-making”?
13. “How do you cite a source from a call with the client’s CEO”?

Appendix B

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